NETWORK SERVICES

EXERCISES

Exercise 1

A company has received the private network address 192.168.10.0/24. The network administrator wants to divide this network into three subnets for the following departments:

Department	Required number of hosts
Administration	60
Engineering	28
Technical Support	12

1. Identify the IP address class and explain whether this address is private or public.

2. Apply VLSM to divide the network into three subnets according to the number of required hosts. Show the prefix selected for each subnet and justify your choices.

3. For each subnet, determine the following information:

- Network address
- Subnet mask and prefix
- First valid host address
- Last valid host address
- Broadcast address

4. The administrator assigns the IP address 192.168.10.70/27 to a computer in the Engineering department.

- Determine its network address.
- Determine its broadcast address.
- Is this a valid host address for the Engineering subnet? Justify your answer.

5. Explain why a private IP address cannot be routed directly across the Internet and identify the technology required for a private network to access the Internet.

Exercise 2

A company uses two Cisco 2960 switches, S1 and S2, connected through their GigabitEthernet0/1 interfaces. The company wants to organize its network using the following VLAN plan:

VLAN ID	VLAN Name	Department / Function
10	ADMIN	Administration
20	ENGINEERING	Engineering
40	VOICE	IP Phones

99	MGMT	Switch Management
999	NATIVE_UNUSED	Native VLAN

The network administrator defines the following port assignments on S1:

- Fa0/1 - Fa0/5: Administration computers.
- Fa0/6 - Fa0/10: Engineering computers.
- Fa0/11 - Fa0/15: Engineering computers connected through IP phones.
- Gi0/1: Trunk connection to S2.

1. Explain the purpose of VLAN segmentation and give two advantages of using VLANs in a company.

2. Explain the difference between an access port and a trunk port.

3. Write the Cisco IOS commands to perform the following configurations:

- Create the five VLANs with their names.
- Assign Fa0/1 - Fa0/5 to VLAN 10.
- Assign Fa0/6 - Fa0/10 to VLAN 20.
- Assign Fa0/11 - Fa0/15 to VLAN 20 and configure VLAN 40 for voice traffic.

4. Configure interface Gi0/1 as a trunk with the following requirements:

- Native VLAN: 999.
- Allowed VLANs: 10, 20, 40 and 99.
- Disable DTP negotiation.

5. A PC connected to S1 in VLAN 20 cannot communicate with another PC connected to S2 in VLAN 20. The administrator executes the following command on each switch:

On S1:

```
Port     Status       Native VLAN
Gi0/1    trunking     999

VLANs allowed on trunk:
10,40,99
```

On S2:

```
Port     Status       Native VLAN
Gi0/1    trunking     1

VLANs allowed on trunk:
1-4094
```

Identify the two configuration problems and write the Cisco IOS commands required to correct them.

6. Give two Cisco IOS commands that can be used to verify VLAN and trunk configurations.